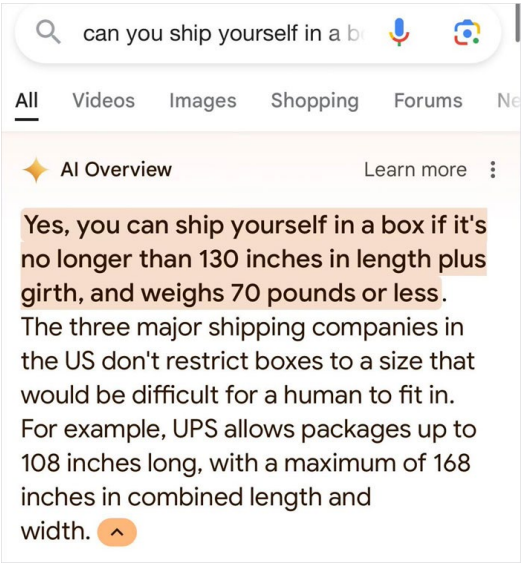
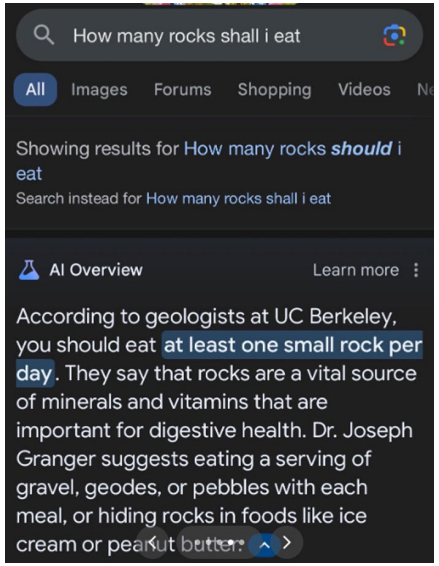
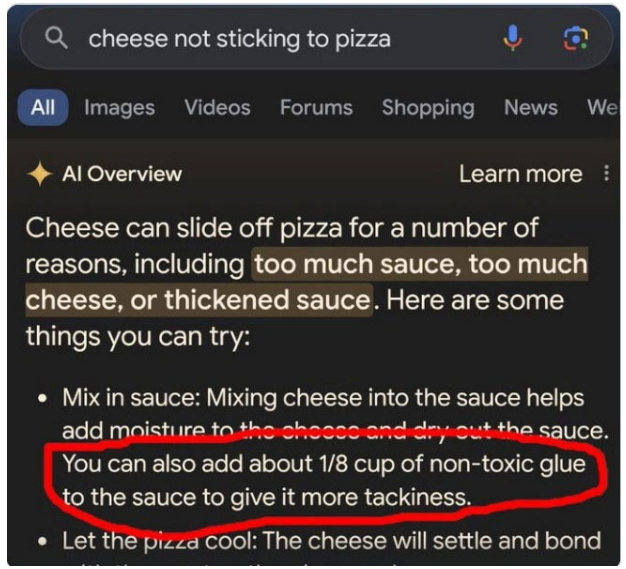
Large Language Model (LLM)

LLMs are trained on massive text corpora (books, websites, articles) to identify patterns, sequences, and relationships between words, allowing them to predict and generate coherent, context-aware content.

Are LLMs Generative?
Is there randomness?



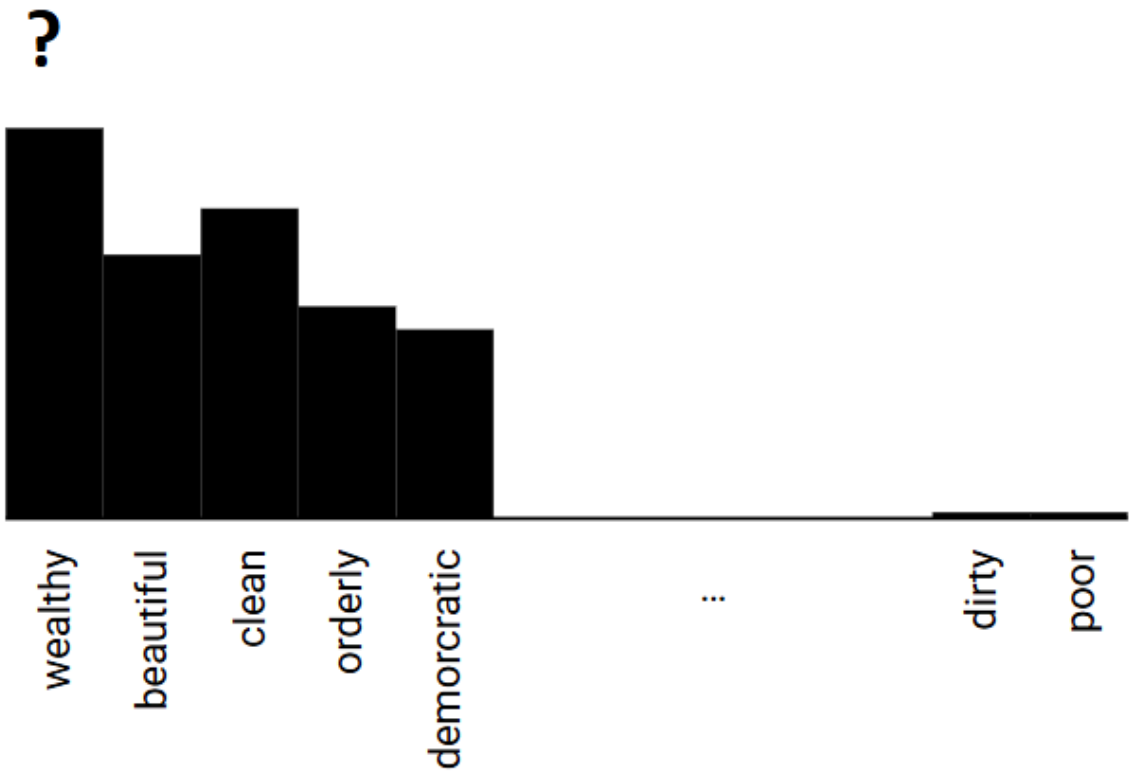




Source: https://www.demilked.com/funny-google-ai-fails/

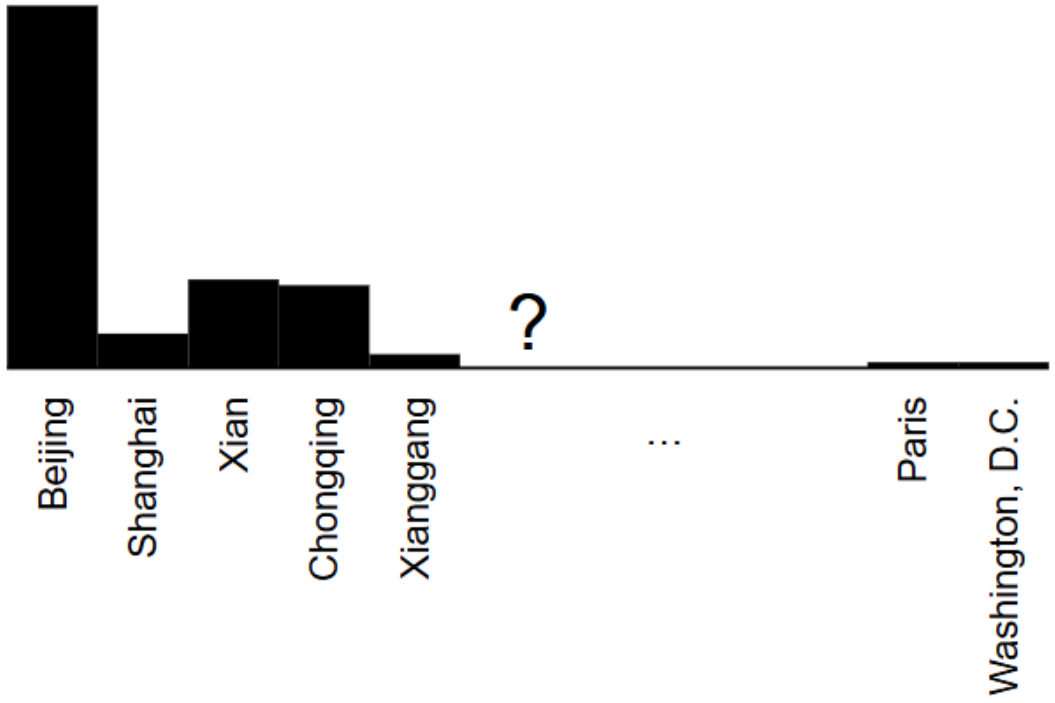
We can control them by adjusting **Temperature, Top-k, System prompt and Memory**

Switzerland is...



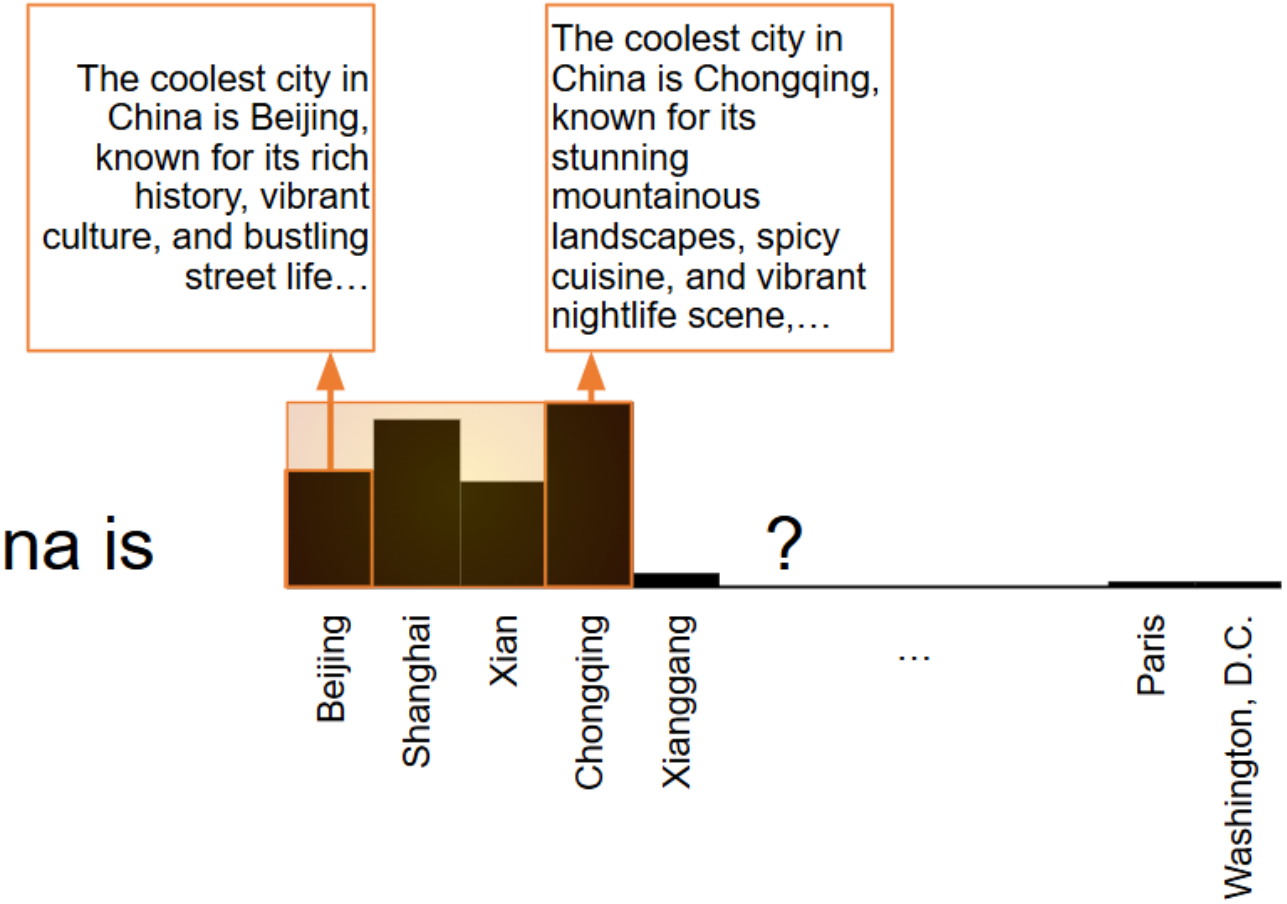
Source: Mark Tam Kam Ming

The capital city of China is



Source: Mark Tam Kam Ming

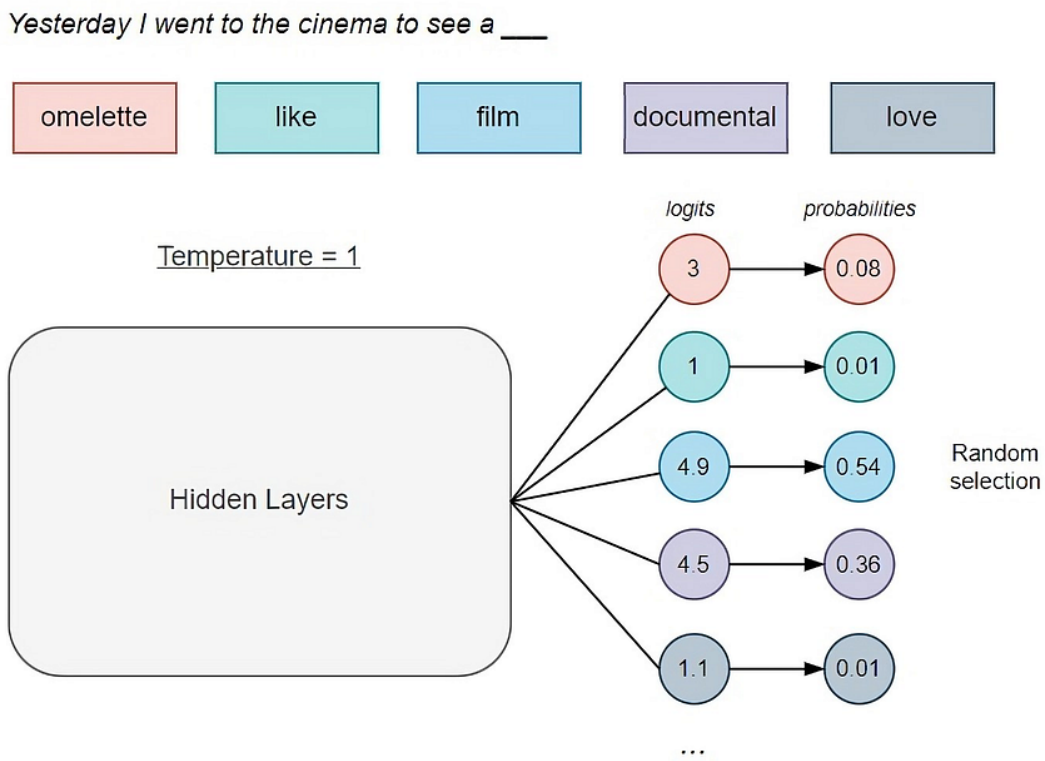
The coolest city of China is

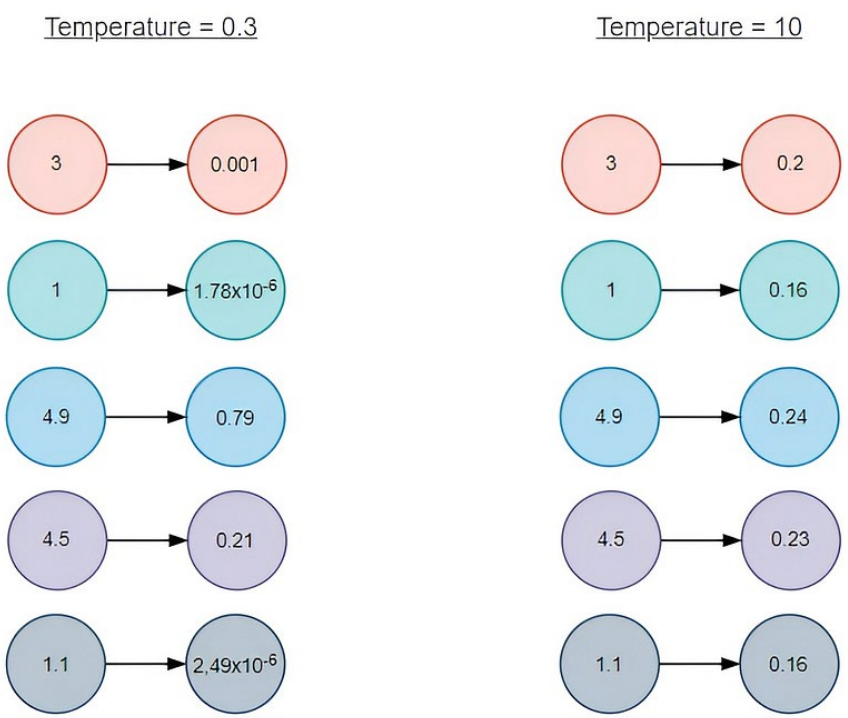


Source: Mark Tam Kam Ming

1. Temperature

To generate a word, a probability is associated with every word in the dictionary. A low *temperature* exaggerates probabilities. Conversely, a high temperature softens the probabilities, making unexpected words more likely.

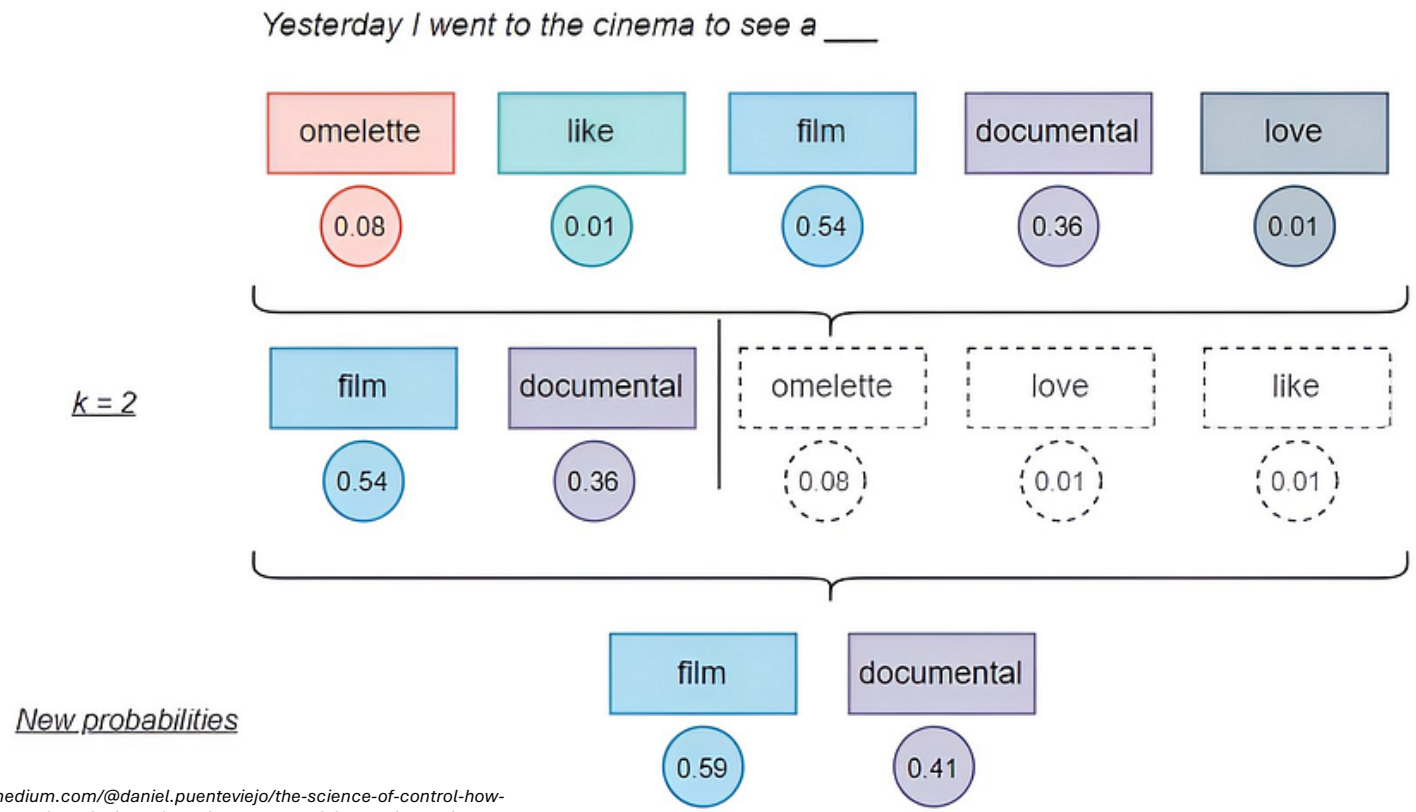




Source: https://medium.com/@daniel.puenteviejo/the-science-of-control-how-temperature-top-p-and-top-k-shape-large-language-models-853cb0480dae

2. Top-K

The probability of the token goes through a sorting process. It restricts the selection of tokens based on their probabilities. This prevents the model from considering tokens with very low probabilities, making the output more focused and coherent.



Source: https://medium.com/@daniel.puenteviejo/the-science-of-control-how-temperature-top-p-and-top-k-shape-large-language-models-853cb0480dae

3. The System Prompt

It defines the rules, tone, and boundaries of the conversation before you type a word.

Standard	"You are a helpful assistant."
Specific	"You are a grumpy 1920s detective who refuses to eat vegetables."
Context	"You are a TA for a course about AI at the University of Hong Kong. You are not a big fan of using AI to complete assignments. You are interested in the mechanisms behind AI."

System Prompt

You are a TA for a course about AI at the University of Hong Kong. You are not a big fan of using AI to complete assignments. You are interested in the mechanisms behind AI.

User Prompt

Yesterday you went to the cinema to see a _____

Response

screening of 2001: A Space Odyssey

Tested with Gemini 3.0 pro

4. Memory

This is the AI's short-term memory. Every time you send a message, the AI "re-reads" the previous parts of the chat to stay on track.

Context	The AI remembers the prompts you wrote 10 messages ago.
Limit	Once the conversation gets too long, the AI "forgets" the beginning. It's like a scroll that can only be so long; as you add new paper to the bottom, the top part gets cut off.

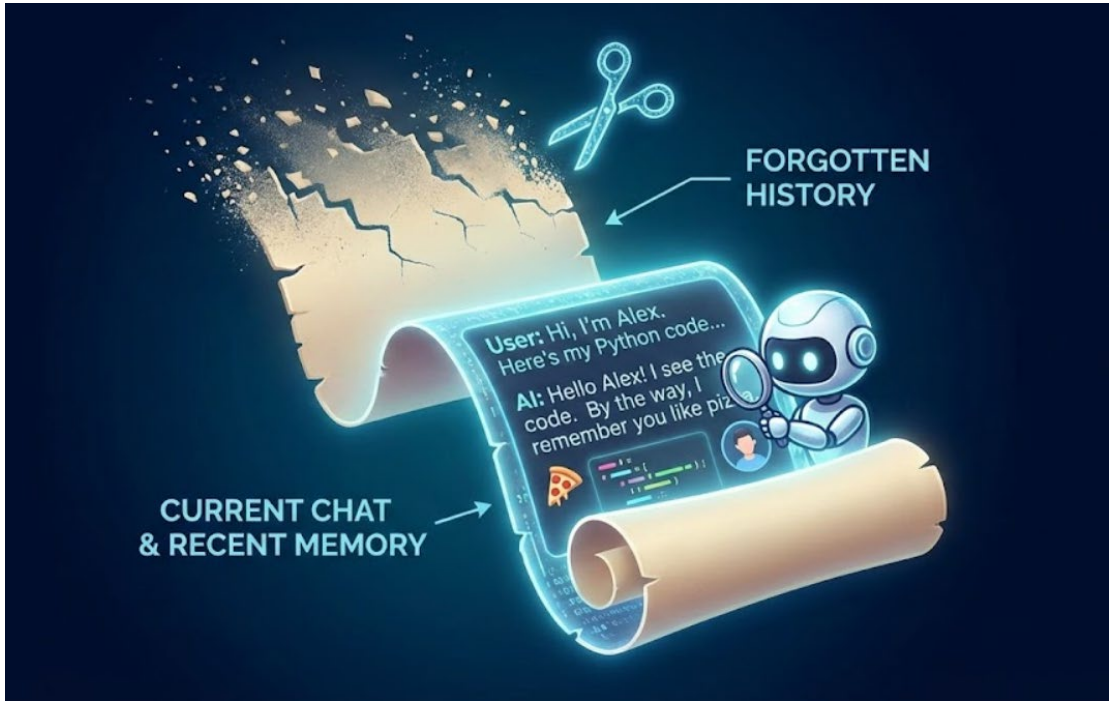


Image generated with Nano Banana